

Age Estimation from Speech: A Data Science Approach

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Abstract—This study explores the use of machine learning to predict speaker’s age based on speech features. By analyzing a data set with a diverse acoustic, linguistic, and demographic attributes, the performance of two models—K-Nearest Neighbors (KNN) and Random Forest—was evaluated. The analysis revealed that Random Forest performed better in capturing relationships within the data, making it more suitable for this task. These findings highlight the role of acoustic features in speaker’s age estimation and point to opportunities for improvement through advanced techniques and additional feature extraction.

I. PROBLEM OVERVIEW

This project aims to develop a system for predicting the age of a speaker based on features extracted from their speech. The data set consists of 3,624 samples, divided into a development set with 2,933 samples and an evaluation set with 691 samples. Each sample corresponds to a spoken sentence, with the speaker’s age provided as the target variable, represented as a continuous numerical value.

The data set includes the following features:

Acoustic Features

- Mean pitch, maximum pitch, and minimum pitch (measured in Hz): Capture variations in the speaker’s vocal frequency.
- Jitter: Measures pitch instability.
- Shimmer: Captures amplitude instability.
- Energy: Represents the overall power of the speech signal.
- Zero-crossing rate (ZCR) mean: Indicates the rate at which the signal changes sign, reflecting tonal changes.
- Spectral centroid mean: Represents the “center of mass” of the frequency spectrum.
- Harmonic-to-noise ratio (HNR): Measures the proportion of harmonic sound to noise, indicating voice quality.
- Duration of silence: Refers to the total time in seconds where no speech is detected.

Linguistic Features

- Number of words: Provides insights into the amount of content spoken.
- Number of characters: Reflects the length of the spoken content.
- Number of pauses: Indicates the speaker’s pause patterns and fluency.

- Tempo: Measured in beats per minute (BPM), it reflects the speaker’s speaking rate.

Demographic Metadata

- Gender: Captures the speaker’s gender information.
- Ethnicity: Captures the speaker’s ethnic background to study potential correlations with speech features and age.

These features were extracted to comprehensively capture the acoustic and linguistic properties of the speech signal, facilitating age prediction. Studies have shown that acoustic characteristics such as pitch, jitter, HNR and energy are strong indicators of the age of a speaker due to physiological changes in the vocal cords over time [1].

In addition to acoustic features, linguistic characteristics such as speech rate (tempo) and demographic metadata, including ethnicity, have been shown to significantly contribute to age estimation accuracy. These features capture not only physiological aspects but also cultural and linguistic variability in speech patterns, which play a critical role in predicting the age of the speaker [2].

Similarly, frequent pauses in speech are significant indicators of speaker’s age, reflecting age-related declines in respiratory and vocal function. Older adults tend to pause more often as a compensatory mechanism, leading to shorter utterances and disrupted linguistic flow. In addition, listeners often associate frequent pauses with characteristics such as frailty or forgetfulness, making pauses both physiologically relevant and socially significant for age estimation [3].

II. PROPOSED APPROACH

A. Preprocessing

The preprocessing step ensures that the data is prepared for modeling by addressing inconsistencies, scaling numerical features, and encoding categorical variables. The following steps were applied:

- 1) Feature Selection: Initially, the data set consisted of 20 columns. Features such as *id*, *path*, and *sampling_rate* were removed for the following reasons:
 - *id*: A unique identifier for each sample, which does not contain meaningful information for predicting age.
 - *sampling_rate*: The sampling rate was the same for all data points, providing no variability.

- *path*: The path indicates the location of the audio file and does not contribute to the prediction task.

After removing these features, the dataset was left with 17 columns relevant for modeling.

- 2) **Data Cleaning**: A typo in the *gender* feature, where "female" was misspelled as "famale" in the evaluation set, was corrected to ensure consistency. Additionally, certain features such as *tempo* were not of the correct data type (e.g., float) and were converted to the appropriate type before normalization.
- 3) **Data Splitting**: The development set was split into a *training set* (80%) and a *test set* (20%) using the train-test-split method to evaluate the model's performance during training.
- 4) **Normalization**: Numerical features were normalized using the *z-score normalization* technique to standardize their distribution. Specifically, the following features were normalized: *mean_pitch*, *max_pitch*, *min_pitch*, *jitter*, *shimmer*, *energy*, *zcr_mean*, *spectral_centroid_mean*, *tempo*, *hmr*, *num_words*, *num_characters*, *num_pauses*, and *silence_duration*. The *fit_transform* method was applied to the training set, while the *transform* method was used for the test and evaluation sets to avoid data leakage.

To illustrate the effect of normalization, a density plot of the *tempo* before and after normalization was created (see Figure 1). This visualization highlights how normalization centers the data around zero and scales it to have a standard deviation of one.

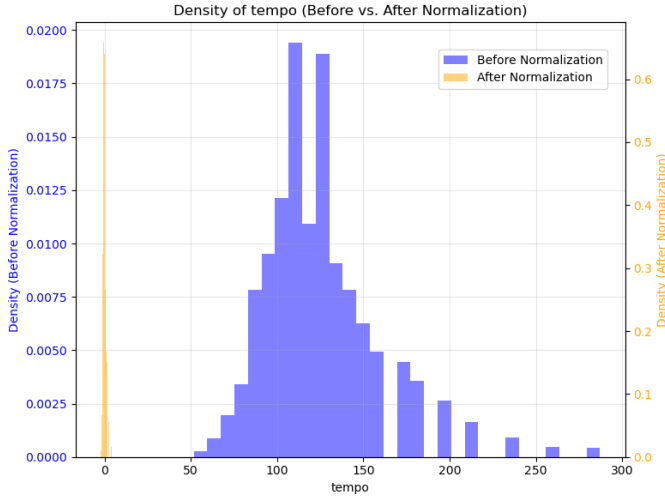


Fig. 1. Density of tempo (Before vs. After Normalization).

- 5) **Categorical Encoding**: The categorical features *gender* and *ethnicity* were encoded using *one-hot encoding*. After encoding, the total number of columns increased to 167.

These preprocessing steps ensured that the data is cleaned, consistent, and prepared for effective model training and evaluation.

B. Model Selection

To identify the most suitable model for predicting the age of the speaker, two machine learning algorithms were evaluated: K-Nearest Neighbors (KNN) and Random Forest.

K-Nearest Neighbors (KNN): K-Nearest Neighbors is a non-parametric, instance-based learning algorithm used for both classification and regression tasks. For regression, KNN predicts the target value by averaging the values of the k -nearest neighbors in the feature space. The key advantages of KNN include its simplicity, lack of assumptions about the underlying data distribution, and flexibility in capturing local relationships in the data. However, KNN can be computationally expensive for large data sets due to its reliance on distance calculations and is sensitive to the choice of k and distance metrics [4].

Random Forest Regressor: Random Forest is an ensemble learning method that constructs multiple decision trees during training and combines their predictions (via averaging for regression) to improve accuracy and robustness. It introduces two key concepts: bagging, where bootstrap samples are used to train individual trees, and feature randomness, where a subset of features is randomly selected at each split to reduce correlation among trees. Random Forest is highly effective for regression tasks due to its ability to handle non-linear relationships, robustness to outliers, and resistance to overfitting. However, it can be less interpretable compared to simpler models [5].

C. Hyperparameter tuning

Hyperparameter tuning was performed for both K-Nearest Neighbors (KNN) and Random Forest Regressor. Different configurations were evaluated for each algorithm to identify the best-performing model.

K-Nearest Neighbors (KNN): Hyperparameter tuning for KNN was conducted by testing different values of k . The goal was to evaluate how the choice of k influences the model's performance on the data set.

Random Forest Regressor: To determine the optimal configuration for the Random Forest Regressor, several combinations of hyperparameters were tested. Key hyperparameters included:

- **Number of Estimators ($n_{estimators}$)**: Determines the number of trees in the forest.
- **Maximum Depth (max_depth)**: Controls the depth of each tree to manage model's complexity.
- **Minimum Samples per Split (min_sample_split)**: Specifies the minimum number of samples required to split a node.

III. RESULTS

K-Nearest Neighbors (KNN): The performance of the K-Nearest Neighbors Regressor was evaluated using the Root Mean Squared Error (RMSE) metric on the test set for different values of k . The results show a gradual improvement in RMSE as k increases, stabilizing after $k = 30$. The best performance on the test set was observed at $k = 30$, with an RMSE of **10.365**.

The RMSE values for the test set are summarized in Table I:

TABLE I
RMSE VALUES FOR DIFFERENT k VALUES (TEST SET)

Number of Neighbors (k)	RMSE
3	11.894
5	11.231
10	10.604
20	10.422
30	10.365
35	10.374
40	10.458

After identifying $k = 30$ as the best-performing model, it was evaluated on the evaluation set to assess its generalization. The RMSE of the evaluation set was **10.539**, indicating consistent performance and good generalization to unseen data.

Random Forest Regressor: The performance of the Random Forest Regressor was evaluated using the Root Mean Squared Error (RMSE) metric for various hyperparameter configurations. The results show a gradual improvement in RMSE as the number of estimators ($n_estimators$) increases, with slight variations when adjusting other parameters like maximum depth (max_depth) and minimum samples per split (min_sample_split). The best performance on the test set was observed with $n_estimators = 500$, $max_depth = 30$, and $min_sample_split = 3$, yielding an RMSE of **9.936**.

The RMSE values for the different configurations on the test set are summarized in Table II:

TABLE II
RMSE VALUES FOR DIFFERENT CONFIGURATIONS (TEST SET)

$n_estimators$	max_depth	min_sample_split	RMSE
100	None	2	10.043
200	None	2	10.003
500	None	2	9.938
500	20	5	9.943
500	30	3	9.936
500	50	3	9.939

After selecting the best-performing configuration on the test set ($n_estimators = 500$, $max_depth = 30$, $min_sample_split = 3$, $random_state = 42$), the model was evaluated on the evaluation set. The RMSE of the evaluation set was **10.215**, indicating slight performance degradation but consistent generalization to unseen data.

Overall, Random Forest outperformed KNN in terms of RMSE on the evaluation set. KNN achieved an RMSE of 10.539, while Random Forest achieved an RMSE of 10.215. These results highlight the better performance of Random Forest, demonstrating better generalization and accuracy for this task.

The feature importance analysis provides insight into the factors contributing to the Random Forest model's predictions. Features such as *silence_duration* and *spectral_centroid_mean* emerged as the most significant predictors, emphasizing the role of temporal and spectral characteristics in speaker age estimation. Furthermore, features such as *jitter*, *num_pauses*, and *hnr* demonstrated

moderate importance, further highlighting the relevance of vocal stability and pauses. In contrast, features such as *gender* and certain ethnicity categories contributed minimally, indicating their limited influence on the model's predictive capability.

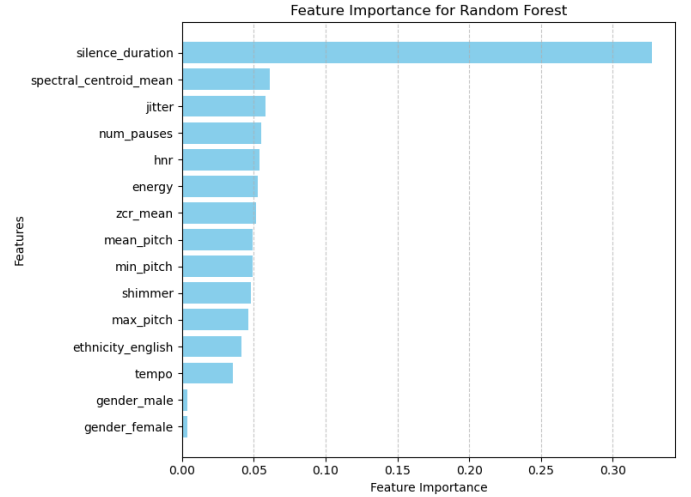


Fig. 2. Feature importance for the Random Forest model. The most significant features include *silence_duration* and *spectral_centroid_mean*, while *gender* and certain ethnicity features contribute minimally.

IV. DISCUSSION

The key findings of this study highlight the better performance of Random Forest over KNN in predicting speaker age. The model successfully identified temporal and spectral features, such as *silence_duration* and *spectral_centroid_mean*, as critical predictors, while demographic features such as *gender* and specific ethnicity categories showed minimal contribution to the predictions.

A limitation of this study lies in the scope of feature extraction. While the features used in the dataset provided meaningful predictions, incorporating additional acoustic parameters such as mean fundamental frequency (*F0 mean*), alpha ratio, and smoothed cepstral peak prominence (*CPPS*) could further enhance model performance by capturing finer aspects of vocal aging [6].

Future work could focus on expanding the feature set and exploring advanced modeling approaches, such as deep learning architectures, to better capture temporal dependencies in speech.

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